

Yufeng Gu

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EDUCATION

University of Michigan

Ph.D. Student, Computer Science and Engineering Department
Advisor: Prof. Reetuparna Das, GPA: 3.88/4.00

Ann Arbor, MI, USA

Sep. 2020 - Aug. 2025

Zhejiang University (ZJU)

B.Eng., College of Information Science and Electronic Engineering
GPA: 3.96/4.00

Hangzhou, China

Sep. 2016 - June 2020

EMPLOYMENT

Intel Labs

Graduate Research Intern, Manager: Nilesch Jain

Optimized AI QoS on next generation heterogenous datacenter platform.

Hillsboro, OR, USA

June 2022 - Aug. 2022

Tenstorrent Inc.

Performance Architect Intern, Manager: Wei-han Lien

Built performance model for RISC-V CPU and AI accelerator.

Santa Clara, CA, USA

May 2023 - Aug. 2023

RESEARCH EXPERIENCE

University of Michigan

Graduate Research Assistant, Advisor: Reetuparna Das

Ann Arbor, MI, USA

Sep. 2020 - Present

GenomicsBench: A Benchmark Suite for Genomics

- Collected time consuming kernels from commonly used genomics secondary analysis pipeline.
- Profiled instruction and memory access characteristics of genomics kernels.

GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis

- Designed the ISA and Microarchitecture of the dynamic programming acceleration accelerator.
- Designed a cycle-accurate simulator to evaluate the accelerator.

AI QoS Optimization on Xeon Server

Yale University

Student Intern, Advisor: James Duncan

New Heaven, CT, USA

Nov. 2019 - Mar. 2020

Interpretation on ASD with fMRI and Deep Learning Models

- Used LSTM-VAE to extract features from Regions of Interests (ROIs) in fMRI.
- Applied graph neural network on interpreting ASD Biomarkers and classification.
- Applied privacy-preserving federated learning framework on multi-site fMRI analysis.

École Polytechnique Fédérale De Lausanne (EPFL)

Student Intern, Advisor: Babak Falsafi

Lausanne, Switzerland

July 2019 - Sep. 2019

QFlex Simulator

- Build CMake compilation for QFlex simulator.
- Patch for GCC-8, Boost Library and Travis-CI code formatting test.

Activation Function Optimization for DNN Accelerator

- Optimized LSTM activation functions with second-order polynomial approximations and lookup

table for both forward computation and back-propagation.

- Designed digital implementation of the optimized activation functions in Verilog.

Zhejiang University (ZJU)

Student Intern, Advisor: Peng Liu

Hangzhou, China

Sep. 2018 – May. 2019

DNN Accelerator

- Designed a systolic array based matrix-matrix multiplication unit.

PUBLICATIONS

Yufeng Gu, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin et al. "GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis." In Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA), pp. 1-15. 2023.

[1] Arun Subramaniyan, **Yufeng Gu**, Timothy Dunn, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, and Reetuparna Das. "GenomicsBench: A Benchmark Suite for Genomics." In 2021 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS), pp. 1-12. IEEE, 2021.

[2] Xiaoxiao Li, **Yufeng Gu**, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. "Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results." Medical image analysis 65: 101765. *IF = 11.148*

[3] Xiaoxiao Li, Yuan Zhou, Nicha C. Dvornek, **Yufeng Gu**, Pamela Ventola, and James S. Duncan. "Efficient Shapley Explanation for Features Importance Estimation Under Uncertainty." In International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 792-801. Springer, Cham, 2020.

SKILLS

- **English Test:** TOEFL 103 (R29/ L26/ S21/ W27), GRE 321 (V151/ Q170/ AW4.0)
- **Programming Language:** C/C++, Python, Verilog, Linux Bash, MATLAB.
- **Development Skills:** VCS Design Compiler, PyTorch, TensorFlow, ModelSim, Vivado, Latex.
- **Others:** Swimming, Tennis, Badminton, Table Tennis, Chinese Calligraphy.

AWARDS & ACHIEVEMENTS

- Fellowship of Summer@EPFL (2% applicants awarded) *July 2019*
- Tang Lixin Fellowship (60/36,000 students in Zhejiang University) *Nov. 2017, 2018, 2019*
- Outstanding Student Leaders in Zhejiang University (3% students awarded) *Oct. 2017, 2019*
- Honorable Mention in Mathematical Contest in Modeling (MCM) *Feb. 2018*
- First-Class Scholarship for Outstanding Students (2% students awarded) *Oct. 2017*

SERVICES

- Artifact evaluation reviewer for IISWC 2021, ISCA 2023.
- Reviewer for University of Michigan graduate student admission, 2021, 2022.
- Voluntary teacher in Tuanlin Primary School, Guizhou *Aug. 2017*